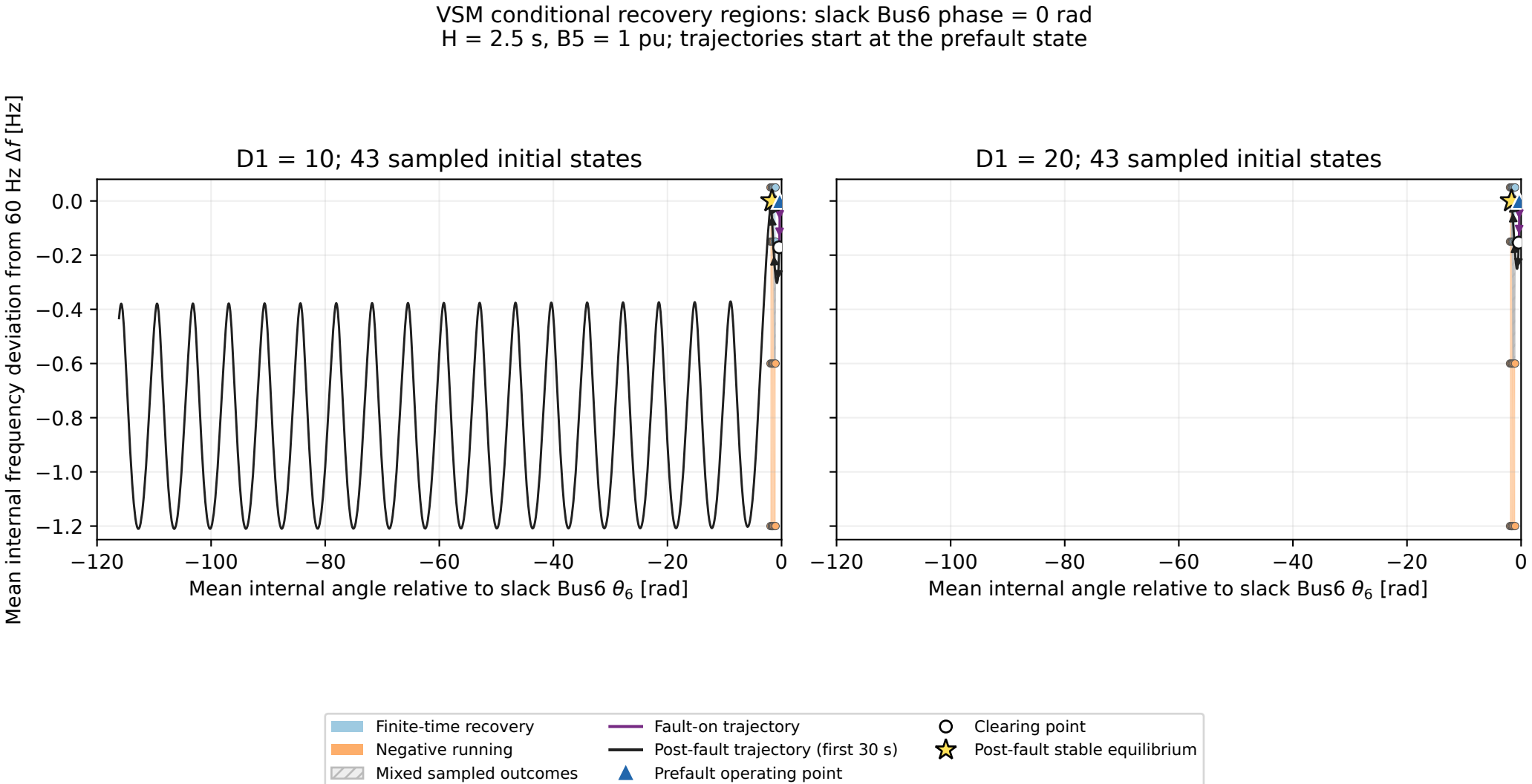
VSM conditional recovery regions: slack Bus6 phase = 0 rad
H = 2.5 s, B5 = 1 pu; trajectories start at the prefault state
D1 = 10; 43 sampled initial states
D1 = 20; 43 sampled initial states
Mean internal frequency deviation from 60 Hz Δf [Hz]
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
−120
−100
−80
−60
−40
−20
0
Mean internal angle relative to slack Bus6 θ6 [rad]
Finite-time recovery
Negative running
Mixed sampled outcomes
Fault-on trajectory
Post-fault trajectory (first 30 s)
Prefault operating point
Clearing point
Post-fault stable equilibrium